



DEALER BEST PRACTICE SERIES

Material Transport
Strategy

Application Maintenance Site Management Component Rebuild Component Life Management Safety

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1.0 Introduction

A parts/component transport strategy is critical to several facets of CRC operations. Transport begins with major component receipt and ends with major component release to the customer, parts department, etc. These activities are non-value adding and are typically unplanned and ad-hoc in nature. Rebuild materials should be moved between processing steps in a manner that consumes minimal resources (time, labor, equipment). Effective part/component moving reduces equipment and staging space requirements and technician rebuild labor.

An effective transport strategy stresses consistency and minimizes “lost” in-process product. Production peaks are less costly and easier to manage due to greater process realization. Properly used, an enhanced material handling strategy provides increased capacity, efficient parts flow, and better labor utilization.

2.0 Best Practice Description

For the scope of this Best Practice, CRC material handling begins with component receipt from the customer at the dealer store. It ends with the component’s release to the customer at the dealer store. The “customer” may also include dealer pick-up and delivery trucks, or dealer field service trucks that transport components.

Current practice is typically ineffective and inconsistent.

- Once the customer arrives with the component, it is often unloaded by rebuild (production) technicians while supervisors “meet and greet” the customer. Unloading and customer communications is often inefficient since many employees (the closest ones to the customer’s component) perform these tasks. These inefficiencies are similar to those occurring when the rebuilt components are loaded-out to the customer.
- Once the components are received, the production employee must move the component from staging, to washing, to disassembly/cleaning, etc., until the rebuilt component is packaged to ship-out.
- During each component move, the employee typically must:
 - Find a lift truck (often in use, or left in random parking places).
 - Find lifting brackets, chains, and straps (often in use, or left in random locations).
 - Find component stands for components brought in, propped-up on cribbing.
 - Interrupt their core duties and resume them later (at the expense of efficiency and quality).

An enhanced material transport strategy is customer friendly and efficient. Generally, the parts department (properly trained and equipped material handlers) must be responsible for almost all heavy material transfers. They are responsible for operating and maintaining transport equipment. Skilled material handlers are trained to use the most effective transport equipment and already transport new, Reman, and exchange/core items. These parts runners continuously travel throughout the dealer’s branch. In fact, “empty” return trips are an efficiency opportunity. All heavy material transfers should be restricted to these material handlers. Allowing the parts department to focus on effective material movement frees the CRC to focus on rebuilding components. The remainder of this description will describe each significant type of material movement and its considerations.

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Parts Dept. and Material Handlers



- All customer receipt and ship functions are handled by the parts department. To receive components: they meet/greet, unload, and stage components. They then transfer components to the proper area per the situation. For example, Reman and exchange cores would have different destinations. Once the component is needed, the parts department transfers it from (protected and organized) staging to the service department wash bay, and then to the disassembly bay. Once the component is rebuilt and sealed, the parts department transfers it to staging and eventually to the customer's vehicle.
- New, Reman, and core items are received between the parts warehouse and specific CRC work bays. This delivery is performed by the material handlers. Technicians do not transfer parts from a central "will-call" or drop area to their work bays.
- Components are moved between work bays as needed for processing. The majority of component moves are performed by material handlers. For example: The handler would transfer sub-components from rebuild bay staging to general assembly bays per a component consist list. Each CRC should be equipped with generic, low-platform push carts (stored under a work bench) for incidental parts moves. This allows for process flexibility with minimal lift truck requirements.
- Miscellaneous items (such as salvage, warranty, or customer hold parts, tools, stands, baskets) are moved from work bay "out" areas, to other storage/ processing areas. These moves may be within a shop, between buildings, or from one branch to another.
- These processes should be utilized for the majority of transports. While minimizing incidental transfers/handling by service technicians, such incidents cannot be prevented entirely.



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- Effective communication is critical for this strategy's success. Technicians and administration must have work bay and office access to call on material handlers. Systems such as in- and out-staging areas, tags, etc., provide for efficient material hand-offs. Tags are also used to organize and direct transport for down-stream processes, such as salvage directions and priorities. One dealer installed "help" lights at bay pick-up/drop-off locations. The lights provided directives to material handlers in the area and were even more effective than first anticipated.

3.0 Implementation Steps

The suggested implementation is as follows:

1. Document as-is material moves, participants, and resources required.
2. Define future transport strategy based on expected product type, volume, size, etc.
 - a. Plan for typical rebuild volumes as well as expected peak production business cycles.
 - b. Investigate specialized transport/handling equipment (radios, intercom, cranes, lift truck attachments, tugs, carts, shelves, chains, racks, etc.).
 - c. Utilize Cat Operations Development and shop equipment supplier expertise as needed.
 - d. Focus on the typical requirements, covering 80-90% of transport circumstances. Add-in flexible capabilities (such as assigned push carts) for minority requirements.
3. Apply future strategy to shop and staging area layout plans.
 - a. Consider the effects of staging location on product protection and flow paths.
 - b. Add changes to shop layout drawing.
 - c. Include shop technician teams in the review/application stages.
 - d. Use Cat Facility Development and Service Operations expertise as required.
4. Develop/document the new material transport strategy/procedure, with new roles and responsibilities.
5. Adjust...
 - a. Repair/rebuild time requirement targets as needed (capture resultant efficiencies).
 - b. Service labor charge-out rates to reflect time and material repair efficiency gains.
 - c. Administrative duties to reflect new system requirements and efficiency gains.
6. Present strategy to shop employees.
7. Implement transport processes, layout, and equipment changes into the shop/storage areas.
 - a. This may occur in phases (as needed).
8. Train employees/customers to use the new procedures/equipment. Follow-up and enforce immediately.
9. Review process/equipment performance once established.
10. Establish and install any adjustments as needed.



Help light installed at a pick-up location indicating material ready to be picked up

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4.0 Benefits

- Increased capacity: Reduced task interruptions, reduced rebuild turnaround times, increased component rebuilds per year. Administration has more time available (with less customer contact) to plan and organize.
- Reduced cost: Fewer moving tools (lift trucks) are needed less often. Related equipment maintenance and repairs are reduced.
- Increase efficiencies: Service technicians focus on core duties reducing labor variances. Rebuild and target times drop, raising standard (flat-rate) job profits.
- Increasing quality: Rebuilt components incur less incidental damage and contamination, while technicians focused on core duties reducing rework.
- Improved Operator/Technician Safety: Trained material handlers increase professionalism and technician-handling problems are eliminated.

5.0 Resources Required

- Equipment Requirement Gaps Vary
 - Most dealers already have more transport equipment than is needed for this strategy.
 - Bay-specific staging racks/equipment are often custom made by the dealer shop.
- Facility
 - Staging space (protected) is located in parts department and each CRC work bay.
- Support Equipment - Cost vary greatly per region
 - Effective cranes provide highest efficiency and shipping/receiving professionalism but lift truck are often the economic choice.
 - Staging racks, carts, stands, and protection for incoming components.
 - Staging racks and protection for out-going components.
 - Protective staging areas for incoming/out-going components.
 - Standard and custom designed carts for bay-to-bay material movement.
 - Communication equipment (radios, intercoms, "help" lights) is low cost but critical.
 - Floor pallets typically consume lots of space and their use is discouraged.
- People
 - Establish current staging concerns with team.
 - Establish process, layout, and equipment improvements through team.
- Training
 - Define, implement, and enforce the new process and duties with the production team.

6.0 Supporting Attachments / References

Component Rebuild Service Operations Development & Assessment (SODA)

7.0 Related Best Practices

[In-Process Staging Strategy](#) [1106-4.4-1038](#)

[Recommended Shipping and Storage Stands for Major Components](#) [0107-4.4-1053](#)

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8.0 Acknowledgements

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